



cycle4green

LENZING
PAPIER



ENVIRONMENTAL FACTS

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PAPER RECYCLING IN EUROPE

Release paper used in labelstock:

Consumption in EU: 310 000 tons (2016)

Recycling rates in EU:

Recycling-%, printing papers	ca. 74 %
Recycled volume, printing paper	ca. 65 Mio tons
Recycling-%, silicone coated paper	15-20 %
Recycled volume, silicone coated paper	54 000 tons

Impact on CO₂ emissions:

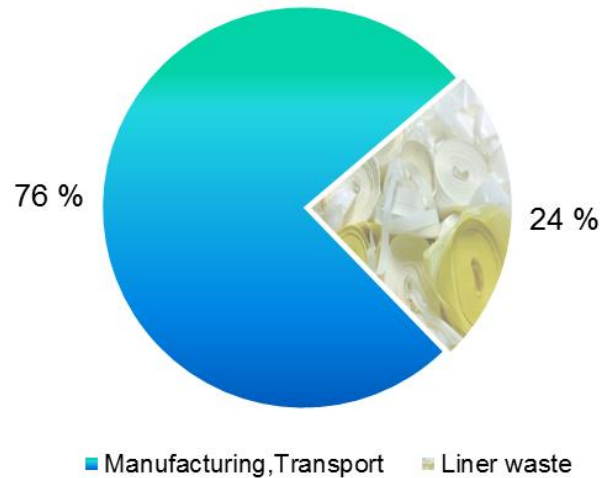
Reduction of emissions :	1,50 ton CO ₂ / ton waste paper (printing)
Reduction of emissions :	2,02 ton CO ₂ / ton waste paper (release)

Impact on water consumption:

Reduction of water usage:	37,5 m ³ / ton pulp
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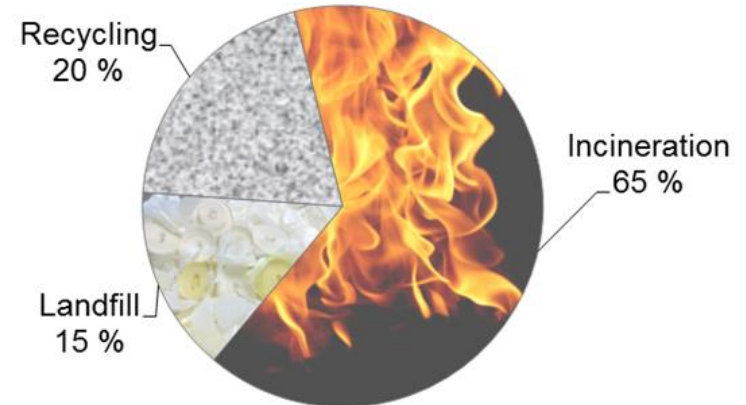
CARBON FOOTPRINT 1/2

Carbon footprint - Self-adhesive label material



- ❑ a quarter of environmental impact originates from liner waste

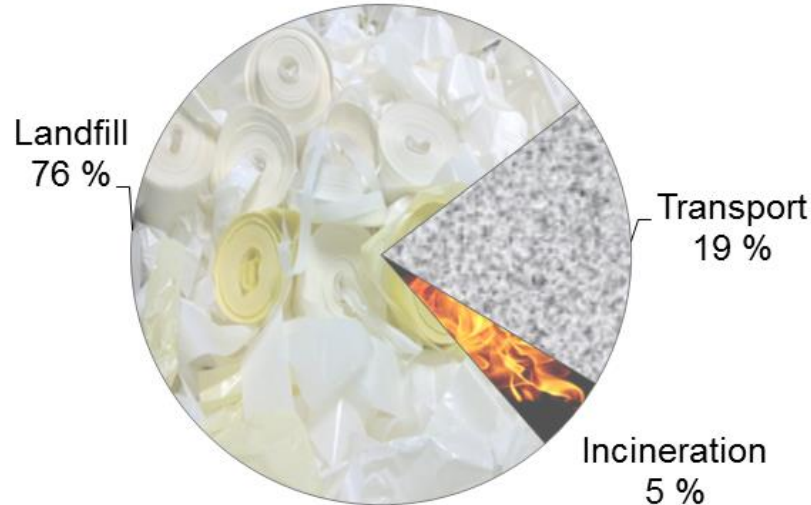
End of life - Release Waste Paper
% of volume



- ❑ Landfilling is becoming less mainly due to more efficient incineration
- ❑ Incineration of organic renewable material is regarded close to carbon neutral providing materials are available locally

CARBON FOOTPRINT 2/2

Emissions - Release Waste Paper % CO₂ ekv.



- ❑ Landfilling is ca. 15% of end-of-life disposal but it dominates emissions as organic material releases methane in landfill
- ❑ Emissions from incineration do not take into account emissions from manufacturing of new products needed after disposal

CONCLUSIONS

- ✓ **Emissions are dominated by land filling**
- ✓ **Recycling locally in Europe brings lowest carbon load**
- ✓ **Potential reduction of emissions -15% by C4G Liner Recycling**

REMARKS:

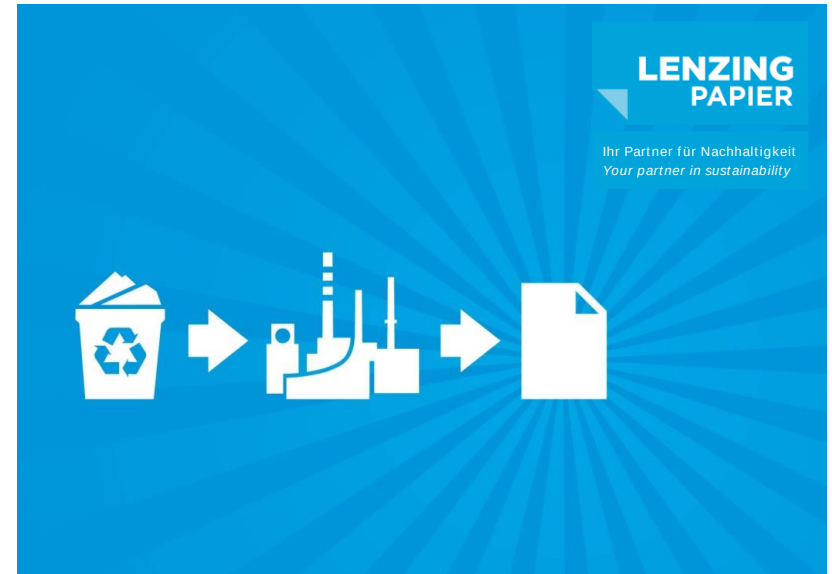
1. Incineration (energy) is carbon neutral provided that raw material production and incineration is done locally without intercontinental transports
2. Transport: recycling circle 750 km, incineration circle 100 km, overseas 9000 km
3. Calculations based on volumes related to C4G recycling

REFERENCES:

1. CEPI Guidelines v1.0 2014
2. GHG Protocoll Transport v1.3
3. CEPI Carbon footprints for paper&boards products Sept. 2007, ENV-148-07
4. IPCC (Intergovernmental Panel on Climate Change) GWP (global warming potential)
5. VTT 2009, 776 Painotuotteen hiilijalan jälki

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